# FIKR Model — Lesson Plan

*Learning Leading to Production*

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| **Title** | Properties of Logarithms | **Subject / Unit** | Topic 6 — Exponential and Logarithmic Functions · Lesson 6-5 |
| **Grade** | Grade 11 — Algebra II | **Date** | Week 7 · Semester 1, 2026–27 |
| **Format** | [ ] 40-minute [ X ] 60-minute | **Teacher** | Mr Malek Thiab |

**Essential Question (curriculum map)**

How can the properties of logarithms, including the change of base formula, be applied to simplify and evaluate logarithmic expressions?

**Lesson Objectives (curriculum map — verbatim)**

1. Apply properties of logarithms to rewrite logarithmic expressions and evaluate logarithms.
2. Use the change of base formula to rewrite and evaluate logarithmic expressions.

**Vocabulary (curriculum map) · Assessments (curriculum map)**

Vocabulary: change of base formula ; logarithmic properties.

Assessments: “Give it a go” questions during practice ; “Time to Check” at the Mastery Gate.

## 0 Preparation Before Class

**Learning Outcome: What will the student be able to do?**

Students will use the product, quotient and power properties of logarithms to expand and condense logarithmic expressions and to evaluate logarithms (objective 1; HSA.SSE.A.2).

Students will use the change of base formula to rewrite and evaluate logarithms, with a calculator and exactly, and will express the solution of $ab^{ct} = d$ as a logarithm and evaluate it (objective 2; HSF.LE.A.4).

Students will use logarithms to find the term number of a geometric sequence given recursively or explicitly (HSF.BF.A.2).

If time runs short, cut Smart Production — never the Mastery Gate, which is the evidence.

**Pre-Class Preparation — Topic/Link and Evidence of Readiness: What will the student complete before class?**

Flipped video (4 min): “Powers add when you multiply” — the exponent laws that become the logarithm properties. Posted on the LMS 48 hours ahead.

Two embedded EdPuzzle questions: write $2^3 \cdot 2^4$ as one power, and evaluate $\log_2 32$.

Evidence of readiness: EdPuzzle completion report plus both embedded answers submitted before the bell.

## 1 Diagnostic 6 min

**Tool Used: What evidence indicates differences in students' current levels?**

3-question Quizizz (≤ 4 minutes) — the slide-4 checks: evaluate $\log_2 32$; write $2^3 \cdot 2^4$ as one power; evaluate $\log_{10} 0.01$.

Answers: $5$ ; $2^7$ ; $-2$. No calculator.

**Where is the learning gap? What will you adjust in your instruction accordingly?**

Expected gap 1 — converting between logarithmic and exponential form (Lesson 6-3); if questions 1 and 3 fail, the properties have nothing to stand on.

Expected gap 2 — Q2: students multiply the exponents instead of adding them; adding is the seed of the product rule.

Routing: 0–1 correct → re-teach converting forms and the exponent laws with the teacher. 2 → straight to the properties. 3 → move quickly through block 1 and begin the Investigate prompt early.

## 2 Targeted Instruction 18 min

**Required Instruction and Modeling: What needs direct explanation? Which example will demonstrate the thinking process and the expected quality standard?**

Slide 5 — Products and quotients (objective 1; HSA.SSE.A.2). $\log_b(MN) = \log_b M + \log_b N$ and $\log_b \frac{M}{N} = \log_b M - \log_b N$. Examples: $\log_2(8x) = 3 + \log_2 x$; $\log_5 \frac{25}{y} = 2 - \log_5 y$; $\log 4 + \log 25 = 2$.

Slide 6 — Powers come to the front (objective 1). $\log_b M^p = p \log_b M$. Examples: expand $\log_5 \frac{x^2\sqrt{y}}{z^3}$; condense $3\log_2 x - \log_2 y = \log_2 \frac{x^3}{y}$.

Slide 7 — The change of base formula (objective 2; HSF.LE.A.4). $\log_b x = \frac{\log_a x}{\log_a b}$. Examples: $\log_5 40 \approx 2.292$; exactly $\log_4 8 = \frac{3}{2}$; solve $3 \cdot 2^t = 100$.

Slide 8 — Logarithms and geometric sequences (HSF.BF.A.2). $n - 1 = \log_r \frac{a_n}{a_1}$. Examples: $a_1 = 3$, $a_{n+1} = 2a_n$ reaches $768$ at $n = 9$; $a_n = 5(1.5)^{n-1}$ first passes $100$ at $n = 9$.

Narration focus — only a product or quotient INSIDE a logarithm splits: $\log(M + N)$ does not.

**Quick Check for Understanding: What question or evidence confirms that students are ready to move on to practice?**

On whiteboards, 60 seconds: condense $\log_2 3 + \log_2 5$.

Expected: $\log_2 15$. Watch for $\log_2 8$ (adding the arguments) and $\log_2 3 \cdot \log_2 5$.

80% correct → release guided practice. Below 80% → one more modelled example with a different base.

## 3 Practice 12 min

**Guided Task: How will the student progress from guided to independent practice? What feedback will the student use?**

Two "Give it a go" problems, identical for every student, worked together with the teacher:

(1) Expand $\log_3 \frac{27}{x}$. (2) Evaluate $\log_2 10$ to 3 decimal places.

Answers: (1) $\log_3 27 - \log_3 x = 3 - \log_3 x$. (2) $\frac{\ln 10}{\ln 2} \approx 3.322$.

Feedback: worked live on the board step by step; students self-mark and circle their OWN first error.

**Independent Task and Feedback Tool: students choose their route and may switch at any time**

Delivered through the "Choose Your Route" sheet with live feedback in Pear Deck. Routes are named for the task, not for the student.

PRACTICE — Secure the method. Expand three expressions, condense $3\log_2 x + \log_2 5$, evaluate $\log_4 30$ and exactly $\log_{27} 9$. Done when: each property is named beside the step that uses it, and every decimal is rounded only at the last step.

APPLY — Use it in context. Find $\log_b 12$ and $\log_b 1.5$ from given values; the first term of $a_n = 4(1.5)^{n-1}$ above $50$; the year a model of Red Sea cruise bookings reaches $4\,000$. Done when: factors are written before any property is used, and the year is a whole number checked in the model.

INVESTIGATE — Find out why. Whether $\log_2(x+4) = \log_2 x + 2$; why $\log_a b \cdot \log_b a = 1$; why LOG and LN give the same value in the change of base formula. Done when: each argument starts from the definition or from numbers, not only from the formula.

Every student sits the same Mastery Gate. The gate is not differentiated, because it is the evidence.

## 4 Production — Stage One 12 min

**Production Task: What new task, situation, or product will allow the student to demonstrate learning?**

Non-routine transfer task — this context has not appeared in practice (slide 10).

A classroom model for yearly visitors to a Red Sea destination on the Jeddah coast: $a_n = 20\,000\,(1.2)^{n-1}$ visitors in year $n$, and each visitor spends a model 1 500 SAR. The model is invented for practice, not a forecast.

(a) Write the first four terms and the recursive rule, and check $a_4$ with the formula. (b) Find the first year with more than 100 000 visitors using a logarithm and the change of base formula. (c) Find the model revenue in that year in millions of SAR.

Answers: (a) $20\,000$, $24\,000$, $28\,800$, $34\,560$; $a_{n+1} = 1.2\,a_n$; $a_4 = 20\,000(1.2)^3 = 34\,560$. (b) $n - 1 > \frac{\ln 5}{\ln 1.2} \approx 8.83$, so $n = 10$ ($a_9 \approx 85\,996$, $a_{10} \approx 103\,196$). (c) $1500 \times 103\,196 \approx 154.8$ million SAR.

Students then hand their reasoning to the AI critic.

**Product Quality Criteria and the Role of AI: What is the minimum acceptable standard? What must the student complete independently?**

Minimum acceptable standard: the exact expression written before any decimal, the year number justified by testing the year before and the year itself, and units (visitors, SAR) in every answer.

Role of AI: critic only. Briefed prompt — "Challenge my reasoning and point out any step I have not justified." Never "solve this".

Completed independently by the student: the written justification for parts (a)–(c). AI may challenge it; it may not write it.

## 5 Presentation — Evidence of Learning 6 min

**Evidence of Learning and Success Criterion: What will the student present or explain, and how? What must be clearly demonstrated for learning to be considered achieved?**

"Time to Check" — one individual task. No notes, no partner, no AI.

1. Condense $2\log_3 x + \log_3 5$. 2. Evaluate $\log_7 50$ to 3 decimal places. 3. Given $\log_b 2 = 0.3$ and $\log_b 5 = 0.7$ (model values), find $\log_b 40$.

Answers: 1. $\log_3(5x^2)$. 2. $\frac{\ln 50}{\ln 7} \approx 2.010$. 3. $3(0.3) + 0.7 = 1.6$.

Done when: each answer names the property used, and question 2 shows the ratio before the decimal.

Scored live in Pear Deck as submissions arrive; result logged in the Impact Dashboard Mastery Rate.

## 6 Final Product — Refinement 6 min

**Refinement and Final Product: What revision will the student make based on feedback? What will the final product look like?**

Students revise their visitor-model reasoning using the AI critique and the teacher's micro-conference note.

Final product, three equally acceptable formats: a Canva one-pager, a photographed hand-written solution, or a 45-second voice note.

Required content: the table of terms, the exact expression for the year, and the year for one new target; ONE worked non-example — writing $\log_{1.2}(5 + x)$ as $\log_{1.2} 5 + \log_{1.2} x$ — with a sentence on why a sum inside does not split; and a caption on why planners test the year before and the year itself.

Graded on originality and clarity of reasoning, not correctness alone.

**Presentation/Publication and Reflection: Where will the product be presented or published? What reflection question will close the learning cycle?**

Published to the class LMS portfolio; the strongest go on the Mathematics Department wall as the Topic 6 modelling set.

Reflection question: "Which is harder for you — choosing the right property, or choosing the right base for the change of base formula?"

**Total minutes across the seven stages:** 60 (6 + 18 + 12 + 12 + 6 + 6) — matches the 60-minute format, including transitions.

## ★ Required Pillars Differentiation · Adaptive Learning · Saudi Connection · Exam Alignment

## Differentiation — three routes named for the task; students choose, and may switch at any time

PRACTICE — Secure the method. Expand, condense and evaluate. Done when: each property is named beside its step.

APPLY — Use it in context. Given-value logarithms, a geometric sequence and a cruise-bookings model. Done when: the year is a whole number checked in the model.

INVESTIGATE — Find out why. A counter-example, the reciprocal pair $\log_a b \cdot \log_b a = 1$, and LOG against LN. Done when: each argument starts from the definition or from numbers.

No student is labelled. The routes describe tasks; every student sits the same Mastery Gate.

## Adaptive Learning — how the lesson branches on evidence

Entry branch — the Quizizz result routes each student to the re-teach table, straight to the new content, or the Investigate prompt.

Mid-lesson branch — Pear Deck flags error patterns during independent practice; flagged students get a 2-minute micro-conference while others continue.

Exit branch — the Mastery Gate routes each student: PASS → Enrichment & Challenge; NOT YET → Targeted Learning Clinic at the start of the next lesson.

Routing is communicated privately through the LMS, never read aloud.

## Saudi Culture & Real-World Connection

Production and project context: a model for yearly visitors to a Red Sea destination on the Jeddah coast (an invented classroom model), with a model spend of 1 500 SAR per visitor — a Vision 2030 tourism-growth setting.

Discussion prompt: why do planners want the year a target is reached, not just the number of visitors in a given year?

## SAT / SAAT (Tahsili) / GAT (Qudurat) Connection

SAAT (Tahsili) — if $\log_2 3 = a$, find $\log_4 12$: change of base gives $\dfrac{a+2}{2}$. Name the trap: forgetting the $+2$ from $\log_2 4$. No calculator.

SAT — Advanced Math: if $2^a = 5$ and $2^b = 3$, find $2^{2a-b} = \dfrac{25}{3}$. Name the trap: subtracting instead of dividing.

GAT (Qudurat) — which is greater, $\log_{10} 25 + \log_{10} 4$ or $2$? They are equal, because the sum is $\log_{10} 100$. Name the trap: adding inside the logarithm.

## Suggestions for Improvement: